

# # FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness

## ## Abstract

Transformers are slow and memory-intensive for long sequences because standard self-attention exhibits  $O(N^2)$  time and memory complexity in sequence length  $N$ . We introduce FlashAttention, an IO-aware exact attention algorithm that uses tiling to reduce the number of memory reads/writes between GPU High Bandwidth Memory (HBM) and GPU on-chip SRAM.

## ## 1. Introduction

Standard attention implementations materialize intermediate  $N \times N$  attention matrices  $S = QK^T$  and  $P = \text{softmax}(S)$  in slow HBM. Because GPU SRAM is significantly faster (19 TB/s) than HBM (1.5 TB/s on A100), memory access is the primary performance bottleneck. FlashAttention computes exact softmax attention without materializing the large intermediate attention matrix.

## ## 2. Algorithm & IO-Aware Tiling

FlashAttention splits query, key, and value matrices into blocks that fit within on-chip SRAM:

1. Tiling: Loops over blocks of  $K$  and  $V$ , loading them into SRAM.
2. Online Softmax: Incrementally computes scaling factors and running softmax sums  $\ell_i$  and  $m_i$  using the online softmax trick:

$$m^{\text{new}} = \max(m, \max(S_j)), \quad \ell^{\text{new}} = e^{m - m^{\text{new}}} \ell + \sum e^{S_j - m^{\text{new}}}$$

3. Block Recomputation in Backward Pass: To save memory during training, intermediate attention matrices are not saved; instead, they are recomputed on-the-fly in SRAM during the backward pass.

## ## 3. Results & Benchmarks

- Speedup: FlashAttention yields 2x to 4x wall-clock speedup on BERT and GPT-2 compared to PyTorch standard attention.
- Memory Footprint: Memory scaling is strictly linear  $O(N)$  with sequence length instead of quadratic  $O(N^2)$ , enabling sequence lengths up to 64k tokens on single GPUs.
- Exactness: Does not approximate attention; numerical results are exact up to floating-point precision.

## ## 4. Limitations

- Hardware Specificity: Kernel requires tuning for specific GPU SRAM configurations (e.g., CUDA tensor cores).
- Non-attention Compute: Only accelerates the self-attention layer; MLP feed-forward blocks remain unaffected.